Roll No. | | | | | | | | | | |

Total No. of Pages :02

Total No. of Questions : 13

B.Pharmacy (Sem.–1)

PHARMACEUTICAL ANALYSIS-I

Subject Code :BP-102T

M.Code :74645

Date of Examination : 10-01-2026

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. Explain briefly :

a) What are Equivalence and End points?

b) What is the pH of a 0.1 N solution of NH_4OH (pKb 4.75)?

c) *"Mohr's method is carried out at pH 6.5 to 8.0"*. Why?

d) What is the role of mercuric acetate in non-aqueous titration of ephedrine HCl?

e) What is a primary standard? Give two examples.

f) What is a co-precipitation? Give an example.

g) Give the principle of limit test for iron.

h) Give chemical name and formula of calomel.

i) What is limiting current?

j) Give balanced chemical equations for titration of sulfanilamide with $NaNO_2$.

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SECTION - B

2. Plot the titration curve for titration between 0.1 M HCl and 0.1 M NH_4OH by calculating pH at various 0, 50, 75, 90, 99, 99.9,100,100.1,101 and 110 percent neutralization. Suggest the most suitable indicator for this titration.

3. Write a detailed account on principle and various steps involved in gravimetric analysis. Describe thegravimetric estimation of Barium as barium sulphate.

4. What is the principle of potentiometry? Write a detailed account on various reference electrodes used in it.

SECTION – C

5. Describe the construction and working of dropping mercury electrode.

6. A 50 ml sample of aqueous solution of oxalic acid requires 20.0 ml of 0.1 N solution of NaOH. Calculate the strength (in % w/v) of oxalic acid in the sample.

7. Define and classify errors. Write a note on different methods to minimize errors.

8. Give method of preparation and standardization of 0.1 M solution of $KMnO_4$. Give balancedchemical equations and general calculations involved in the method.

9. What is the principle of complexometric titrations? Describe the method for estimation of calcium gluconate. https://www.pyqonline.com

10. Write a detailed account on Fajan's method.

11. Describe the application of conductometric titrations in different types of neutralization titrations.

12. Write a descriptive note on bromatometry.

13. Define the terms Common ion effect and Digestion with example.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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